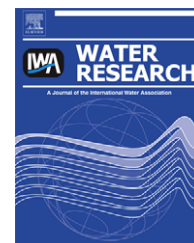


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Review

Present limitations and future prospects of stable isotope methods for nitrate source identification in surface- and groundwater

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ABSTRACT

Nitrate (NO_3^-) contamination of surface- and groundwater is an environmental problem in many regions of the world with intensive agriculture and high population densities. Knowledge of the sources of NO_3^- contamination in water is important for better management of water quality. Stable nitrogen ($\delta^{15}\text{N}$) and oxygen ($\delta^{18}\text{O}$) isotope data of NO_3^- have been frequently used to identify NO_3^- sources in water. This review summarizes typical $\delta^{15}\text{N}$ - and $\delta^{18}\text{O}$ - NO_3^- ranges of known NO_3^- sources, interprets constraints and future outlooks to quantify NO_3^- sources, and describes three analytical techniques (“ion-exchange method”, “bacterial denitrification method”, and “cadmium reduction method”) for $\delta^{15}\text{N}$ - and $\delta^{18}\text{O}$ - NO_3^- determination. Isotopic data can provide evidence for the presence of dominant NO_3^- sources. However, quantification, including uncertainty assessment, is lacking when multiple NO_3^- sources are present. Moreover, fractionation processes are often ignored, but may largely constrain the accuracy of NO_3^- source identification. These problems can be overcome if (1) NO_3^- isotopic data are combined with co-migrating discriminators of NO_3^- sources (e.g. ^{11}B), which are not affected by transformation processes, (2) contributions of different NO_3^- sources can be quantified via linear mixing models (e.g. SIAR), and (3) precise, accurate and high throughput isotope analytical techniques become available.

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1. Introduction

Nitrate (NO_3^-) contamination in water is an environmental problem worldwide. Increased input of reactive nitrogen (N) is attributed to intensive land use, increased use of N-containing organic and inorganic fertilizers (Smil, 1999), animal manure, discharge of human sewage and elevated atmospheric N deposition (Benkovitz et al., 1996). The World Health Organization has set a limit of $10 \text{ mg NO}_3^- \cdot \text{NL}^{-1}$ for drinking water. The implementation of the Nitrate Directive (EC, 2002) in Europe established a detailed framework for prevention of NO_3^- pollution to waters. However, to effectively control NO_3^- contamination in water, NO_3^- source inputs should be better understood. Subsequently, dedicated measures could be applied to prevent or minimize contamination.

Stable N isotope data of NO_3^- ($\delta^{15}\text{N}$ - NO_3^-) have been frequently used to estimate the origin of NO_3^- in water (Wells and Krothe, 1989; Feast et al., 1998; Mayer et al., 2002), because the isotopic composition of N in NO_3^- is generally different among various NO_3^- sources such as atmospheric N_2 , soil, chemical fertilizers, and manure. However, the origin of NO_3^- must be linked to the entire N cycle, since values of $\delta^{15}\text{N}$ - NO_3^- can be biased due to mixing of distinct NO_3^- sources and kinetic isotopic fractionation (e.g. denitrification) (Kellman and Hillaire-Marcel, 1998). Hence, the $\delta^{15}\text{N}$ signature alone does not allow for conclusive identification of NO_3^- sources. Therefore, a dual isotope approach (combination of ^{15}N and ^{18}O) has been widely used and provides more conclusive information for tracing sources of NO_3^- in water (Komor, 1997; Aravena and Robertson, 1998; Widory et al., 2004; Seiler, 2005). In addition, the analytical methodologies for both $\delta^{15}\text{N}$ - and $\delta^{18}\text{O}$ - NO_3^- have been improved considerably in recent years and are ready to become high throughput analytical techniques.

This paper is composed of four parts: (1) basic information on N and O isotopes and isotopic ranges of known NO_3^- sources, (2) factors affecting the isotopic composition of NO_3^- sources, (3) outlooks to improve the identification of NO_3^- sources, and (4) description and evaluation of analytical techniques for $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ determination in NO_3^- . Thus, the overall objective is to provide a comprehensive overview of the current state-of-the-art to identify NO_3^- sources in water via $\delta^{15}\text{N}$ - and $\delta^{18}\text{O}$ - NO_3^- data and to provide an outlook for future improvements for NO_3^- source identification.

2. Stable nitrogen and oxygen isotopes and kinetic isotope fractionation

There are two naturally occurring stable isotopes of nitrogen (N), ^{14}N and ^{15}N . The majority of N in the atmosphere is

composed of ^{14}N (99.6337%) and the remainder is composed of ^{15}N (0.3663%) (Junk and Svec, 1958). Oxygen (O) is composed of three stable isotopes, ^{16}O (99.759%), ^{17}O (0.037%), and ^{18}O (0.204%) (Cook and Lauer, 1968). Stable isotope ratios are usually expressed in delta (δ) units and a per mil (‰) notation relative to the respective international standards:

$$\delta_{\text{sample}} (\text{‰}) = [(R_{\text{sample}} - R_{\text{standard}}) / R_{\text{standard}}] \times 1000 \quad (1)$$

where R is the $^{15}\text{N}/^{14}\text{N}$ or $^{18}\text{O}/^{16}\text{O}$ ratio of the sample and standard, respectively for $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$. $\delta^{15}\text{N}$ values are reported relative to atmospheric air (AIR) and $\delta^{18}\text{O}$ values are reported relative to Vienna Standard Mean Ocean Water (VSMOW). When δ_{sample} is positive, it indicates enrichment in the heavy isotope. A negative δ_{sample} indicates depletion in the heavy isotope.

Stable isotopes are subjected to kinetic isotope fractionation, which results in depleted instantaneous products, while the remaining substrate becomes gradually more enriched with the heavy isotope. The latter is characterized by an enrichment factor (ϵ) in ‰ which is defined as:

$$\epsilon = 10^3 (\alpha - 1) \quad (2)$$

in which α is the fractionation factor and is defined as

$$\alpha = K_{15\text{N}} / K_{14\text{N}} \quad \text{OR} \quad \alpha = K_{18\text{O}} / K_{16\text{O}} \quad (3)$$

where κ are the rate constants in the transformation process. In a closed system, isotopic enrichment can be expressed by the following Rayleigh equation:

$$\delta_{\text{S}(t)} = \delta_{\text{S}0} + \epsilon \ln(S_t / S_0) \quad (4)$$

where ϵ is the isotopic enrichment factor, $\delta_{\text{S}0}$ and $\delta_{\text{S}(t)}$ represent the isotopic compositions of the substrate at time 0 and t, and S_0 and S_t are the concentrations of the substrate at time 0 and t, respectively. This equation can be used to estimate the enrichment factor ϵ by plotting measured values of $\delta_{\text{S}(t)}$ against $\ln(S_t / S_0)$.

3. $\delta^{15}\text{N}$ values of NO_3^- sources

The use of $\delta^{15}\text{N}$ - NO_3^- for identification of NO_3^- sources dates back to the 1970s. The first study using $\delta^{15}\text{N}$ - NO_3^- was reported by Kohl et al. (1971) for estimation of fertilizer contribution to NO_3^- in the Sangamon River (Illinois, USA). Different NO_3^- sources can be discriminated from each other as NO_3^- originating from different sources shows characteristic $\delta^{15}\text{N}$ values. Box plots of $\delta^{15}\text{N}$ data for various N sources and sinks are displayed in Fig. 1, which shows percentile values and outliers of $\delta^{15}\text{N}$ for each NO_3^- source. The typical $\delta^{15}\text{N}$ ranges are presented by the 10th and 90th percentiles.

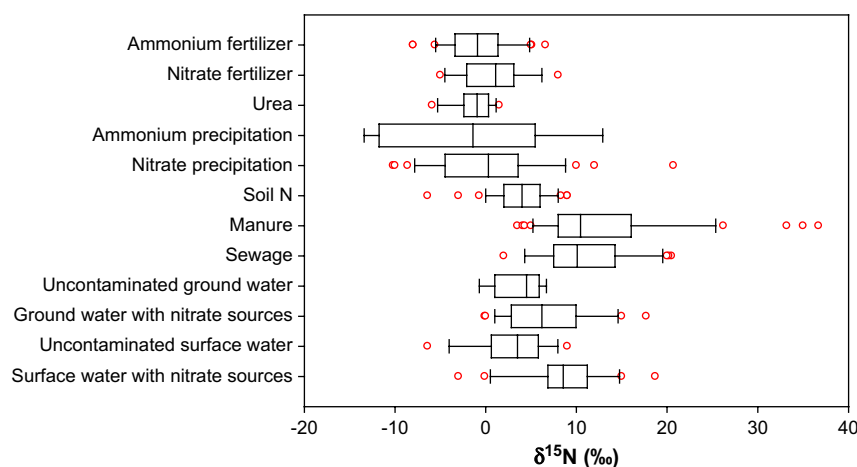


Fig. 1 – Box plots of $\delta^{15}\text{N}$ values of NO_3^- from various sources and sinks. Box plots illustrate the 25th, 50th and 75th percentiles; the whiskers indicate the 10th and 90th percentiles; and the circles represent outliers. NH_4^+ fertilizer – Heaton (1986); Hübner (1986); Cao et al. (1991); Wassenaar (1995); Kendall (1998); Roadcap et al. (2001); Choi et al. (2003b); Singleton et al. (2007); Bateman and Kelly (2007); Choi et al. (2007); Li et al. (2007). NO_3^- fertilizer – Black and Waring (1977); Hübner (1986); Hirata (1996); Kendall (1998); Roadcap et al. (2001); Deutsch et al. (2006a); Bateman and Kelly (2007); Singleton et al. (2007). Urea – Black and Waring (1977); Heaton (1986); Cao et al. (1991); Wassenaar (1995); Choi et al. (2002a); Deutsch et al. (2006a); Bateman and Kelly (2007); Choi et al. (2007); Li et al. (2007). NH_4^+ precipitation – Hübner (1986); Hirata (1996); Russell et al. (1998); Li et al. (2007); Zhang et al. (2008). NO_3^- precipitation – Kendall (1998); Russell et al. (1998); Spoelstra et al. (2001); Williard et al. (2001); Pardo et al. (2004); Kellman (2005); Piatek et al. (2005); Deutsch et al. (2006a); Finlay et al. (2007); Hales et al. (2007); Li et al. (2007); Singleton et al. (2007); Townsend-Small et al. (2007); Lee et al. (2008); Zhang et al. (2008). Soil N – Heaton (1986); Mariotti et al. (1988); Wassenaar (1995); Lindau et al. (1997); Fogg et al. (1998); Kendall (1998); McClelland and Valiela (1998); Mayer et al. (2001); Williard et al. (2001); Kellman (2005); Spoelstra et al. (2007); Singleton et al. (2007). Manure – Wassenaar (1995); Fogg et al. (1998); Kendall (1998); Karr et al. (2001); Choi et al. (2002b); Choi et al. (2003a); Choi et al. (2003b); Curt et al. (2004); Widory et al. (2004); Kellman (2005); Widory et al. (2005); Choi et al. (2007). Sewage – Hirata (1996); Fogg et al. (1998); Kendall (1998); Mariotti et al. (1988); McClelland and Valiela (1998); Curt et al. (2004); Widory et al. (2004); Widory et al. (2005); Li et al. (2007). Uncontaminated groundwater – Burg and Heaton (1998); Brandes et al. (1998); Rock and Mayer (2002); Seiler (2005); Deutsch et al. (2006a); Choi et al. (2007). Groundwater with nitrate sources – Iqbal et al. (1997); Fogg et al. (1998); Mengis et al. (1999); Panno et al. (2001); Min et al. (2002); Choi et al. (2002a); Mitchell et al. (2003); Katz et al. (2004); Jun et al. (2005); Li et al. (2007); Choi et al. (2007). Uncontaminated surface water – Spoelstra et al. (2001); Williard et al. (2001); Mayer et al. (2002); Pardo et al. (2004); Piatek et al. (2005); Deutsch et al. (2006b); Hales et al. (2007); Lee et al. (2008). Surface water with nitrate sources – Rock and Mayer (2002); Pardo et al. (2004); Kellman (2005); Deutsch et al. (2006a); Deutsch et al. (2006b); Voss et al. (2006); Lee et al. (2008).

Ammonium fertilizer, NO_3^- fertilizer and urea are (in)organic fertilizers, which are produced by fixation of atmospheric N_2 and show small differences in $\delta^{15}\text{N}$ content as a result of small fractionation during subsequent processing of the fixed N (Flipse and Bonner, 1985). These (in)organic fertilizers show a typical $\delta^{15}\text{N}$ value range (10th and 90th percentiles in box plot) between -6‰ and $+6\text{‰}$.

The typical $\delta^{15}\text{N}$ values for atmospheric N deposition are situated between -13‰ and $+13\text{‰}$. This is controlled by complex chemical reactions in the atmosphere and various anthropogenic sources such as combustion of fossil fuels (Hübner, 1986; Kendall, 1998).

Manure and sewage are enriched in ^{15}N relative to other N sources. During storage, treatment and application of sewage and animal wastes, ammonia (NH_3) volatilization causes a large enrichment of ^{15}N in the residual NH_4^+ . This NH_4^+ is subsequently converted into ^{15}N -enriched NO_3^- . Thus, $\delta^{15}\text{N}$ values of NO_3^- originating from manure are between $+5\text{‰}$ and $+25\text{‰}$ and sewage between $+4\text{‰}$ and $+19\text{‰}$.

The typical $\delta^{15}\text{N}$ values of soil N range from 0‰ to $+8\text{‰}$. This is linked to the relative rate of mineralization and nitrification. Other factors including soil depth, vegetation, climate and site history may also affect the $\delta^{15}\text{N}$ values of soil (Kendall, 1998; Mayer et al., 2001).

Nitrogen contamination in surface- and groundwater (recognized as NO_3^- sinks) in agricultural regions is generally attributed to the application of fertilizers, animal waste and other organic residues. Additionally, intensive land use, vegetation and industrial activity are also important contributors of NO_3^- to surface- and groundwater. Therefore, the $\delta^{15}\text{N}$ values of NO_3^- in surface- and groundwater typically vary from -4‰ to $+15\text{‰}$.

4. $\delta^{18}\text{O}$ values of NO_3^- sources

Values of $\delta^{18}\text{O}$ - NO_3^- have increasingly been used as an additional means to identify sources of NO_3^- in water (Kendall et al., 1996; Mayer et al., 2002; Pardo et al., 2004; Deutsch et al.,

2006a). Durka et al. (1994) found that $\delta^{18}\text{O}$ was more useful than $\delta^{15}\text{N}$ to separate atmospheric NO_3^- deposition from microbially-produced soil NO_3^- , because $\delta^{18}\text{O}$ signatures of atmospheric NO_3^- (from +52.5‰ to +60.9‰) and microbially-produced soil NO_3^- (from +0.8‰ to +5.8‰) differ significantly. Wassenaar (1995) also suggested that $\delta^{18}\text{O}$ may be a good tracer of NO_3^- sources, because it distinguishes synthetic NO_3^- fertilizers from other NO_3^- sources.

Box plots of $\delta^{18}\text{O}$ signatures of NO_3^- sources are displayed in Fig. 2. It is clear that typical $\delta^{18}\text{O}$ values of NO_3^- from nitrification (including $\delta^{18}\text{O}$ values from microbial production of NH_4^+ in fertilizer and precipitation, NO_3^- derived from soil N and NO_3^- derived from manure and sewage) are lower than that of NO_3^- from precipitation and NO_3^- in fertilizer.

The $\delta^{18}\text{O}$ of microbially-produced NO_3^- , NO_3^- in precipitation and NO_3^- fertilizer are regulated by different processes. The $\delta^{18}\text{O}$ of microbially-produced NO_3^- is determined by the $\delta^{18}\text{O}$ of H_2O and atmospheric O_2 . In theory, two oxygen atoms of the newly formed NO_3^- are derived from H_2O oxygen, with the third oxygen atom incorporated from atmospheric O_2 (Hollocher, 1984; Anderson and Levine, 1986; Böhlke et al., 1997; Durka et al., 1994; Kendall, 1998; Wassenaar, 1995; Mayer et al., 2001). Nitrate derived from nitrification should theoretically have $\delta^{18}\text{O}$ values between –10‰ and +10‰, since environmental water typically has $\delta^{18}\text{O}$ values between

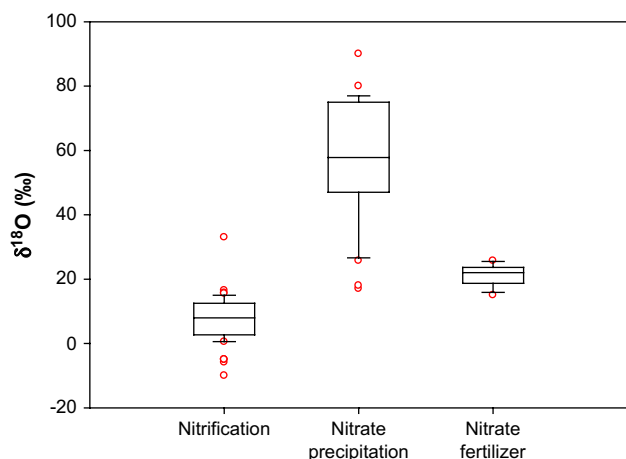


Fig. 2 – Box plots of $\delta^{18}\text{O}$ values of NO_3^- generated during nitrification, nitrate precipitation and nitrate fertilizer. Box plots illustrate the 25th, 50th and 75th percentile; the whiskers indicate the 10th and 90th percentiles; and the circles represent data outliers. Nitrification – Hübner (1986); Wassenaar (1995); Kendall (1998); Mayer et al. (2001); Spoelstra et al. (2001); Williard et al. (2001); Mayer et al. (2002); Rock and Mayer (2002); Fukada et al. (2003); Pardo et al. (2004); Piatek et al. (2005); Seiler (2005); Deutsch et al. (2006a); Hales et al. (2007); Singleton et al. (2007); Spoelstra et al. (2007); Lee et al. (2008); Umezawa et al. (2008). Nitrate precipitation – Kendall (1998); Spoelstra et al. (2001); Williard et al. (2001); Pardo et al. (2004); Piatek et al. (2005); Deutsch et al. (2006a); Finlay et al. (2007); Hales et al. (2007); Singleton et al. (2007); Lee et al. (2008). Nitrate fertilizer – Hübner (1986); Kendall (1998); Roadcap et al. (2001); Deutsch et al. (2006a); Singleton et al. (2007).

–25‰ and +4‰ and $\delta^{18}\text{O}$ of atmospheric O_2 is +23.5‰ (Hollocher, 1984; Durka et al., 1994; Kendall, 1998). It has, however, been observed that $\delta^{18}\text{O}$ values of microbially-produced NO_3^- are up to 5‰ higher than the calculated theoretical maximum of 10‰ (Aravena et al., 1993; Kendall, 1998; Mayer et al., 2001). This can be expected if (1) the $\delta^{18}\text{O}$ value of H_2O is isotopically enriched as a result of evaporation, (2) the $\delta^{18}\text{O}$ value of soil O_2 is higher than that of atmospheric O_2 resulting from O isotope fractionation during respiration, (3) there is significant isotope fractionation during the incorporation of oxygen from H_2O and O_2 into the newly formed NO_3^- , (4) the ratio of oxygen incorporation from H_2O and O_2 is not 2:1 (e.g. more O_2 may be derived from atmospheric O_2 when NH_4^+ is limiting), or (5) a different bacterial process dominates the nitrification reaction and utilizes a greater amount of atmospheric O_2 in low pH environments (Aravena et al., 1993; Wassenaar, 1995; Böhlke et al., 1997; Kendall, 1998; Mayer et al., 2001). Hence, $\delta^{18}\text{O}$ - NO_3^- produced is relatively enriched compared to the theoretical calculation for nitrification.

The $\delta^{18}\text{O}$ - NO_3^- of precipitation is controlled by complex atmospheric processes, which result in large spatial and temporal variability in $\delta^{18}\text{O}$ - NO_3^- . Affecting factors include isotopic fractionation during NO_3^- formation caused by thunderstorms, the isotopic signature of the reactive oxygen in the atmosphere that combines with NO_x to form NO_3^- , and any isotopic fractionation during reactions in the atmosphere (Kendall, 1998; Pardo et al., 2004). Thus, $\delta^{18}\text{O}$ of atmospheric NO_3^- is enriched relative to atmospheric O_2 (+23.5‰), showing a wide range between +25‰ and +75‰.

Synthetic NO_3^- fertilizers have distinctive $\delta^{18}\text{O}$ values (Amberger and Schmidt, 1987), with a range from +17‰ to +25‰, as the O is mainly derived from atmospheric O_2 ($\delta^{18}\text{O} = 23.5$ ‰).

5. Effect of $\delta^{18}\text{O}$ on NO_3^- source identification

Since $\delta^{18}\text{O}$ - NO_3^- is used as an additional marker of NO_3^- sources, it is meaningful to assess the effect size of $\delta^{18}\text{O}$ - NO_3^- data on $\delta^{15}\text{N}$ - NO_3^- for NO_3^- source identification. An effect size is a measure of the magnitude of the relationship between two variables. The calculation of the effect size is carried out by the d -index of Hedges implemented in the Meta Win software (Rosenberg et al., 2000). It is calculated as:

$$d = \frac{\bar{X}_e - \bar{X}_c}{S} J \quad (5)$$

where $\bar{X}_e$ is the mean of the experimental group ($\delta^{18}\text{O}$ - NO_3^-), $\bar{X}_c$ is the mean of the control group ($\delta^{15}\text{N}$ - NO_3^-), S is the pooled standard deviation, and J is a factor that corrects for small sample bias. An effect size value (Hedges' d) equal to zero indicates no effect, and effect sizes of 0.2, 0.5 and 0.8 or greater indicate small, medium and large effects, respectively. Values below zero indicate negative effects.

The effect size of $\delta^{18}\text{O}$ - NO_3^- data on $\delta^{15}\text{N}$ - NO_3^- for NO_3^- source identification is shown in Fig. 3. It is clear that some effect size values are above zero, indicating a positive contribution of $\delta^{18}\text{O}$ - NO_3^- data for NO_3^- source identification, but in some cases effect size values are below 0 (e.g. manure and

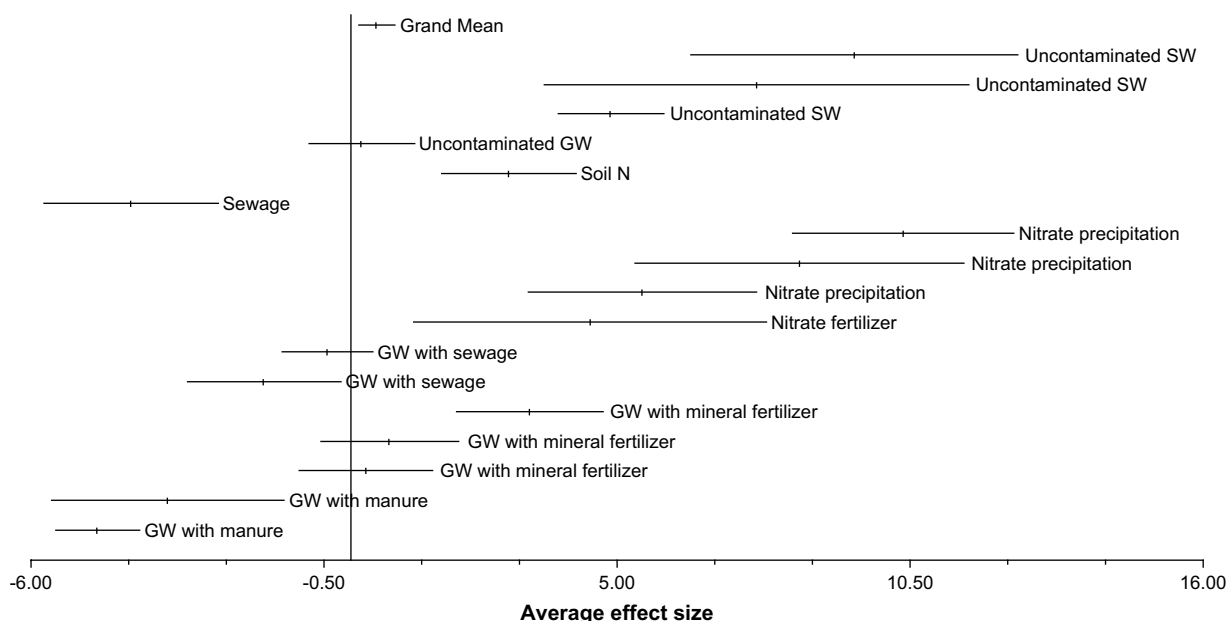


Fig. 3 – Average effect size and 95% confidence intervals of $\delta^{18}\text{O}$ on $\delta^{15}\text{N}$ for NO_3^- source identification. GW represents groundwater; SW represents surface water. GW with manure – Wassenaar (1995); Rock and Mayer (2002). GW with mineral fertilizer – Panno et al. (2001); Deutsch et al. (2006a); Moore et al. (2006). GW with sewage – Fukada et al. (2004); Moore et al. (2006). Nitrate fertilizer – Deutsch et al. (2005). Nitrate precipitation – Pardo et al. (2004); Deutsch et al. (2006a); Hales et al. (2007). Sewage – Aravena and Robertson (1998). Soil N – Hales et al. (2007). Uncontaminated GW – Moore et al. (2006). Uncontaminated SW – Karr et al. (2001); Pardo et al. (2004); Hales et al. (2007).

sewage), indicating negative contribution of $\delta^{18}\text{O}$ - NO_3^- data for NO_3^- source identification. The overall grand mean effect size is 0.5 and the 95% confidence interval [0.13–0.83] does not overlap with zero. Thus, overall $\delta^{18}\text{O}$ - NO_3^- data provide a significant contribution for NO_3^- source identification.

6. Factors influencing isotopic compositions of NO_3^- sources

Multiple NO_3^- sources from agricultural and urban activities contributing to surface- and groundwater and complex fractionations caused by multiple N-cycling processes change the original $\delta^{15}\text{N}$ - and $\delta^{18}\text{O}$ - NO_3^- values, potentially biasing identification of NO_3^- contaminated sources (Kellman, 2005; Kendall, 1998).

Several soil N processes control the $\delta^{15}\text{N}$ values of both the NH_4^+ available to be nitrified and the formed NO_3^- . The ammonification process (organic-N $\rightarrow$ NH_4^+) results in a small fractionation ($\pm 1\text{‰}$) between (soil) organic matter and (soil) NH_4^+ (Kendall, 1998). In contrast, the conversion of NH_4^+ to NO_2^- and NO_3^- is accompanied by marked N isotope fractionation effects, resulting in NO_3^- depleted in ^{15}N relative to the $\delta^{15}\text{N}$ of the initial NH_4^+ , while the residual NH_4^+ is characterized by progressively increasing $\delta^{15}\text{N}$ values (Delwiche and Steyn, 1970; Mariotti et al., 1981; Macko and Ostrom, 1994). The enrichment factors are between -12‰ and -29‰ (Kendall, 1998). During nitrification, the $\delta^{18}\text{O}$ of newly produced NO_3^- obviously does not depend on the isotopic composition of the organic matter being mineralized, but depends on the $\delta^{18}\text{O}$ of H_2O and O_2 incorporation (see above).

However, there is emerging evidence that O can exchange between H_2O and intermediate compounds of nitrification (Andersson et al., 1982; DiSpirito and Hooper, 1986; Kool et al., 2007). There are generally three steps in nitrification: firstly, NH_4^+ is oxidized to NH_2OH catalyzed by ammonia monooxygenase, in which O from O_2 is incorporated; secondly, NH_2OH is oxidized to NO_2^- catalyzed by hydroxylamine oxidoreductase, in which O from H_2O is incorporated; and finally, NO_2^- is oxidized to NO_3^- catalyzed by nitrite oxidoreductase with incorporation of O from H_2O , in which O can exchange between H_2O and NO_2^- or NO_3^- as a result of reversibility of this step (Kool et al., 2007).

Another possible process causing significant alterations of the isotopic composition of NO_3^- is microbial denitrification, which occurs when O is limited and organic carbon is available (Knowles, 1982). Kinetic isotope effects during microbial denitrification are responsible for preferentially converting the lighter isotopes ^{14}N and ^{16}O to N_2 and N_2O , causing an enrichment of the heavy isotopes in the remaining NO_3^- (Mariotti et al., 1981; Mayer et al., 2002; Fukada et al., 2003). Nitrogen isotope enrichment factors of denitrification are within -40‰ to -5‰ (Hübner, 1986; Smith et al., 1991; Sebilo et al., 2003), and oxygen isotopic enrichment factors are between -18‰ and -8‰ (Böttcher et al., 1990; Mengis et al., 1999; Fukada et al., 2003; Lehmann et al., 2003). Some studies reported that a linear relationship indicating an enrichment of ^{15}N relative to ^{18}O by a factor between 1.3:1 and 2.1:1 gives evidence for denitrification (Aravena and Robertson, 1998; Mengis et al., 1999; Fukada et al., 2003).

Ammonia volatilization is a process that causes a large enrichment of the heavier ^{15}N isotope in the residual NH_4^+ .

Volatilization involves equilibrium (reversible) reactions and kinetic (irreversible) reactions leading to fractionation. The equilibrium fractionation occurs between NH_4^+ and NH_3 in solution, and between aqueous and gaseous NH_3 . The hydrolysis of urea (the main N form in animal waste) or NH_4^+ fertilizers causes a temporary increase in pH, which favors the NH_3 gas loss by volatilization. The kinetic fractionation is caused by the diffusive loss of ^{15}N -depleted NH_3 , which results in NH_4^+ enriched in ^{15}N (Kendall, 1998).

Mixing of multiple NO_3^- sources is another bias that affects NO_3^- source identification. Although different NO_3^- sources have a distinctive isotopic composition, the mixture shows intermediate values. Moreover, NO_3^- sources in the environment rarely keep the initial composition as various fractionation processes may alter the initial composition before or after mixing (Kendall, 1998). This situation may give rise to incorrect calculations of the contribution of NO_3^- sources in mixing models and inaccurate results for NO_3^- source identification (Panno et al., 2006).

7. Source identification by combining isotope methods with hydrochemistry

Recently, researchers have attempted to link $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ values with land-use types or physico-chemical properties of water to identify NO_3^- sources (Karr et al., 2001; Mayer et al. 2002; Spruill et al. 2002; Mitchell et al. 2003; Kaushal et al., 2006). From three years of observation in rural areas, Choi et al. (2007) reported the following $\delta^{15}\text{N}$ values of groundwater contaminated with NO_3^- from different land-use activities: +4.5‰ to +8.5‰ for cropping areas with mineral fertilizer use, from +8.7‰ to +17.6‰ for farmland with animal manure application, and above +10‰ for residential areas with sewage discharge. Mayer et al. (2002) and Voss et al. (2006) found a significant positive linear relationship ($r^2 = 0.75$, $p = 0.001$, $n = 16$ in Mayer et al. (2002); $r^2 = 0.77$, $p < 0.001$, $n = 11$ in Voss et al. (2006)) between $\delta^{15}\text{N}$ values of riverine NO_3^- and percentages of agricultural and urban land-use. They showed that concentrations and N isotope values of riverine NO_3^- under agricultural plus urban land were markedly higher than in predominantly forested areas with soil N and atmospheric deposition as the main NO_3^- sources. Thus, evidence from the combination of isotopic composition values with land-use types indicates that anthropogenic activities are the predominant NO_3^- sources for contaminated surface- and groundwater.

The combination of concentration and isotopic composition of NO_3^- can also provide useful evidence for identification of NO_3^- sources (Kendall, 1998; Mayer et al. 2002; Pardo et al. 2004). Min et al. (2002) and Choi et al. (2007) found a correlation between increasing NO_3^- concentrations in groundwater and more positive $\delta^{15}\text{N}$ - NO_3^- values when manure was the predominant NO_3^- source. The contribution of manure raised the $\delta^{15}\text{N}$ - NO_3^- signatures above the background levels of soil-derived NO_3^- in groundwater. In contrast, Iqbal et al. (1997) observed a correlation between increasing NO_3^- concentrations in groundwater and more negative $\delta^{15}\text{N}$ - NO_3^- values when fertilizer was the predominant source. The contribution of fertilizers decreased the $\delta^{15}\text{N}$ - NO_3^- signatures below the

background levels of soil-derived NO_3^- in groundwater. However, these correlations could only be functional in the case of a single NO_3^- contamination source and when no fractionation process occurs.

Linking $\delta^{15}\text{N}$ - NO_3^- values with concentrations of anions and cations in water also provides valuable information for identifying NO_3^- sources, as chemical constituents from anthropogenic sources are inclined to dominate those of natural sources (Spruill et al., 2002). Min et al. (2002) performed a study on NO_3^- sources from agricultural practices for river-side alluvial aquifers along the Nakdong River in Korea, and found that based on $\delta^{15}\text{N}$ - NO_3^- values and physico-chemical data (e.g. Ca^{2+} , Na^+ , K^+ , Cl^- , HCO_3^- , etc.) fertilizers, animal manure and human wastes were dominant NO_3^- sources. They reported that when the groundwater type changes from a Ca- HCO_3^- type (background or less-polluted) to a Na(-K)-Cl type, local contamination by domestic sewage or manure might be responsible. Changes of the major anions from HCO_3^- to Cl^- are possibly caused by agricultural activities. Karr et al. (2001) investigated the effects of swine-manure contamination on groundwater using $\delta^{15}\text{N}$ - NO_3^- together with K^+ , Cl^- and NO_3^- data. They pointed out that the source of Cl^- and K^+ enrichment was swine-manure and found positive correlations between NO_3^- , Cl^- , and K^+ in groundwater. They determined the median $\delta^{15}\text{N}$ - NO_3^- of groundwater to be +15.4‰, falling well in the range of manure (+5‰ and +25‰). Thus, demonstrating that swine-manure can be successfully traced. Spruill et al. (2002) built a classification tree model using N isotopes, anions and cations to correctly classify five NO_3^- sources (fertilizer on crops, fertilizer on golf courses, irrigation spray from hog wastes, and leachate from poultry litter and septic systems) in groundwater and obtained a correct classification rate of more than 80%.

8. Boron isotope application for NO_3^- source identification

Apart from land-use types and hydrochemistry, boron isotopes (B) have also been used for the identification of NO_3^- sources in water (Bassett et al., 1995; Vengosh et al., 1999; Widory et al., 2004; Widory et al., 2005). Boron has two stable isotopes (^{11}B and ^{10}B) with natural abundances of approximately 80% and 20%. Boron is highly soluble in aqueous solutions and is present as a minor or trace constituent in nearly all water types (Bassett et al., 1995; Chetelat and Gaillardet, 2005). Furthermore, B is not affected by transformation processes (Vengosh et al., 1994; Bassett et al., 1995). Boron has already been used in several studies as a tracer for sewage contamination in groundwater (Vengosh et al., 1994; Bassett et al., 1995; Eisenhut and Heumann, 1997; Verstraeten et al., 2005) and surface water (Vengosh et al., 1999; Chetelat and Gaillardet, 2005). Values of $\delta^{11}\text{B}$ of sewage reported in the literature range from -7.7‰ to +12.9‰ (Gellenbeck, 1994; Vengosh et al., 1994; Bassett et al., 1995; Leenhouts et al., 1998; Vengosh et al., 1999; Widory et al., 2004; Widory et al., 2005; Seiler, 2005). Animal manure and mineral N fertilizers also contain B as a minor or trace element. The $\delta^{11}\text{B}$ values reported for different types of manure range from +6.9‰ to +42.1‰, and are predominately higher than the $\delta^{11}\text{B}$ values reported

for fertilizers (ranging from +8‰ to +17‰) and the $\delta^{11}\text{B}$ values of sewage. These distinct $\delta^{11}\text{B}$ signatures have been used to discriminate between animal manure sources and sewage or, to a lesser extent, fertilizer sources (Komor, 1997; Widory et al., 2004; Widory et al., 2005). Widory et al. (2004, 2005) demonstrated the benefits of the combined use of $\delta^{15}\text{N}$ and $\delta^{11}\text{B}$ values to identify multiple NO_3^- sources (fertilizers, greenhouse discharges, sewage, hog, cattle and poultry manure) in groundwater in areas with different hydro-geological conditions in France. Seiler (2005) used $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ values combined with $\delta^{11}\text{B}$ values to discriminate between domestic wastewater and fertilizer contamination in groundwater in Nevada (USA).

The main processes which may affect the B isotopic composition are adsorption–desorption interactions with clay minerals, iron and aluminum oxide surfaces and organic matter (Palmer et al., 1987; Bassett, 1990; Goldberg et al., 2000; Yingkai and Lan, 2001; Lemarchand et al., 2005) and mineral precipitation (only in extremely saline environments) (Bassett et al., 1995).

9. Quantification of NO_3^- source inputs

Deutsch et al. (2006a) successfully used the dual isotope approach to quantify three NO_3^- sources (water from artificially drained agricultural soils, groundwater and atmospheric deposition). They also determined the contribution of each NO_3^- source to a river sub-basin in Germany based on a mass-balance mixing model (Phillips and Koch, 2002). Results from the mixing model indicated that NO_3^- from the drainage water contributed 86% and the other two NO_3^- sources contributed 11% (groundwater) and 3% (atmospheric deposition) respectively. The dual isotope approach was also applied to quantify NO_3^- inputs into 12 Baltic rivers by Voss et al. (2006). Similarly, a mass-balance mixing model was used to quantify three major N source contributions, which are sewage, atmospheric deposition and pristine soils.

Normally, a basic mass-balance mixing model based on two isotopes and three sources is expressed as follows:

$$\delta^{15}\text{N}_M = f_1\delta^{15}\text{N}_1 + f_2\delta^{15}\text{N}_2 + f_3\delta^{15}\text{N}_3 \quad (6)$$

$$\delta^{18}\text{O}_M = f_1\delta^{18}\text{O}_1 + f_2\delta^{18}\text{O}_2 + f_3\delta^{18}\text{O}_3 \quad (7)$$

$$1 = f_1 + f_2 + f_3 \quad (8)$$

where $\delta^{15}\text{N}_M$ and $\delta^{18}\text{O}_M$ are the NO_3^- isotope values from the mixture, and the subscripts 1, 2 and 3 represent any of the three NO_3^- sources; f is defined as the proportional contribution of the respective source.

However, Moore and Semmens (2008) mentioned that this mixing model does not take into account the incorporation of several substantial sources of uncertainty. The first source of uncertainty is temporal and spatial variability in $\delta^{15}\text{N}$ - and $\delta^{18}\text{O}$ - NO_3^- in NO_3^- ; the second source of uncertainty is isotopic fractionation during denitrification; and the third is that too many NO_3^- sources contribute to the mixture so that the above equations cannot be solved exactly. To fully incorporate those sources of uncertainty, Parnell and Jackson (2008) have

developed and implemented a stable isotope mixing model called SIAR (<http://cran.r-project.org/web/packages/siar/index.html>), which is a recently released open source software package that runs in R statistical computing program. This model has been successfully applied in food-web analysis, such as estimates of prey (source) contributions to the predator (mixture). SIAR uses a Bayesian framework to establish a logical prior distribution based on Dirichlet distribution (Evans et al., 2000) for estimating possible proportional source contribution, and then to determine the probability distribution for the proportional contribution of each source to the mixture. By defining a set of N mixture measurements on J isotopes with K source contributors, the mixing model which takes the above mentioned uncertainties into account can be expressed as follows (Jackson et al., in press):

$$\begin{aligned} X_{ij} &= \sum_{k=1}^K p_k (S_{jk} + c_{jk}) + \varepsilon_{ij} \\ S_{jk} &\sim N(\mu_{jk}, \omega_{jk}^2) \\ c_{jk} &\sim N(\lambda_{jk}, \tau_{jk}^2) \\ \varepsilon_{ij} &\sim N(0, \sigma_j^2) \end{aligned} \quad (9)$$

where X_{ij} is the isotope value j of the mixture i , in which $i = 1, 2, 3, \dots, N$ and $j = 1, 2, 3, \dots, J$; S_{jk} is the source value k on isotope j ($k = 1, 2, 3, \dots, K$) and is normally distributed with mean μ and standard deviation ω ; p_k is the proportion of source k which needs to be estimated by the SIAR model; c_{jk} is the fractionation factor for isotope j on source k and is normally distributed with mean λ and standard deviation τ ; and ε_{ij} is the residual error representing the additional unquantified variation between individual mixtures and is normally distributed with mean = 0 and standard deviation.

Although SIAR has never been used for quantifying NO_3^- sources, it can potentially provide reliable estimations of different NO_3^- source contributions to surface- and groundwater.

10. Analytical techniques for the determination of ^{15}N - NO_3^- and ^{18}O - NO_3^-

Precise, accurate, but also inexpensive and fast analysis of NO_3^- for both $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ is needed for improved NO_3^- source identification, quantification and uncertainty assessment.

In recent years, the so called “ion-exchange” or “ AgNO_3 method” for both $\delta^{15}\text{N}$ - NO_3^- and $\delta^{18}\text{O}$ - NO_3^- analysis has been developed by Chang et al. (1999) and Silva et al. (2000). This method is used to concentrate and purify NO_3^- in water samples for simultaneous ^{15}N and ^{18}O determination. Briefly, NO_3^- is purified and concentrated by passing samples through cation and then anion exchange resin columns. NO_3^- is eluted using hydrochloric acid, neutralized with silver oxide and then filtered to remove the AgCl . For accurate $\delta^{18}\text{O}$ - NO_3^- analysis, all non-nitrate oxygen-bearing anions (e.g. SO_4^{2-} , CO_3^{2-} , and PO_4^{3-}) are removed from the sample by adding BaCl_2 to the AgNO_3 solution and then the precipitate is filtered out. The filtered solution is then passed through a cation exchange resin to remove excess Ba^{2+} ions and re-neutralized with Ag_2O . The resulting solution can be freeze-dried (Silva et al.,

2000) or oven-dried (Fukada et al., 2003) to produce AgNO_3 salts. $\delta^{15}\text{N}$ analysis of the prepared AgNO_3 can be conducted by conversion to N_2 gas for IRMS analysis via mixing AgNO_3 with a CuO , Cu wire and CaO catalyst and combusting in a sealed tube at 850°C (Kendall and Grim, 1990). $\delta^{18}\text{O}$ analysis can be conducted using the combustion method which generates CO_2 by adding finely ground spectrographic graphite or by using pyrolysis systems that can generate CO by addition of graphite. Alternatively, $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ can be simultaneously analyzed via TC/EA-IRMS (thermal conversion/elemental analyzer-isotope ratio mass spectrometer). Prepared AgNO_3 samples are converted (N_2 and CO) at 1400°C in a molybdenum-lined, aluminum oxide combustion tube which is filled with glassy carbon and topped with a glassy carbon crucible. Upon pyrolysis of the AgNO_3 , the reaction gases are separated via a 1 m E3030 GC column (Elemental Microanalysis) and analyzed via IRMS. The “ion-exchange method” has the following advantages: (1) the concentration of NO_3^- from water samples onto anion exchange resin columns can be accomplished in the field and is convenient for transporting and storing water samples for NO_3^- isotopic analysis; (2) there is minimal isotopic fractionation for NO_3^- stored on anion exchange columns (Silva et al., 2000); and (3) the technique achieves a high level of sensitivity. The disadvantages of the “ion-exchange method” can be summarized as follows: (1) the sample preparation procedure is relatively labor-intensive (3–5 days for sample preparation) and cost-intensive (up to 60 Euro per sample just for consumables only); (2) high concentrations of anions (e.g. Cl^- , SO_4^{2-} , DOC, etc.) in water samples can interfere with the adsorption of NO_3^- onto anion exchange resins; and (3) target sample size of 100–200 μmol of NO_3^- for optimal analysis requires large sample volumes for low NO_3^- concentration samples.

Another sample preparation technique for both $\delta^{15}\text{N}$ - NO_3^- and $\delta^{18}\text{O}$ - NO_3^- determination is the so called “bacterial denitrification method” (Sigman et al., 2001; Casciotti et al., 2002; Rock and Ellert, 2007). This method allows for the simultaneous determination of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ of N_2O produced from the conversion of NO_3^- by denitrifying bacteria which naturally lack N_2O -reductase activity (the enzyme that catalyses the reduction of N_2O to N_2). In brief, bacterial cultures are grown for 6–10 days in amended tryptic soy broth (TSB), divided into centrifuge tubes of 40 mL aliquots and centrifuged. After centrifugation, the supernatant is decanted, reserved and 4 mL of the TSB is pipetted back into the tubes to obtain a 10-fold concentration of bacteria. These tubes are then vortexed to ensure homogenized cultures and then transferred as 2×2 mL aliquots into 20 mL headspace vials. The vials are crimp-sealed with Teflon-backed silicone septa. To ensure anaerobic conditions, a reduced blank effect and removal of N_2O produced prior to sample injection, the headspace vials are purged with N_2 gas for 3 h. Samples of dissolved NO_3^- (100 nmol) are then injected into the headspace vials and are incubated overnight to allow for complete conversion of NO_3^- to N_2O . The next day, 0.1 mL of 10 N NaOH is injected into the headspace vials to stop bacterial activity and to scrub any CO_2 gas in the vial which can interfere with the N_2O measurement. $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ analysis of the produced N_2O can be conducted by (1) extracting N_2O offline via freezing it out in a vacuum line (cryo-concentration) and analyzing by

IRMS (Sigman et al., 2001), or (2) extracting N_2O online via an autosampler injection system and using Nafion as an additional water trap, and then analyzing by IRMS (Casciotti et al., 2002). The “bacterial denitrification method” has the following advantages: (1) the preparation procedure is less labor-intensive (2–3 days for sample preparation) and less expensive (up to 5 Euro per sample for consumables); (2) the technique requires a smaller sample size (three orders of magnitude smaller than the “ion-exchange method”) and allows for the analysis of low NO_3^- concentration samples; and (3) the method achieves a high level of sensitivity. The disadvantages of the “bacterial denitrification method” are summarized as follows: (1) bacterial growth takes a longer time of about 10–12 days (from Petri plate to media bottles); (2) bacterial cultivation is potentially affected by toxicity of the sample (e.g. antibiotics, heavy metal, pesticides, etc.), which is difficult to predict; (3) the presence of NO_2^- in water samples can bias the isotopic composition of product N_2O ; and (4) mathematical approaches are needed to quantify and correct for both fractionation of oxygen isotopes during oxygen atom loss in the reaction process and exchange of oxygen atoms in nitrite and nitric oxide with water that are inherent to the denitrifier method for $\delta^{18}\text{O}$ analysis.

Besides the now well established bacterial conversion of NO_3^- to N_2O and subsequent measurement of $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$, a new method is gaining ground; conversion of NO_3^- to N_2O by a two-step chemical reduction procedure. McIlvin and Altabet (2005) published the “Cadmium reduction method” or “Azide method” which first converts NO_3^- to NO_2^- using cadmium reduction followed by a subsequent reduction to N_2O using sodium azide. The resulting N_2O is then analyzed for $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ in the same way described for the denitrifier method. The first conversion step of this technique (reduction of NO_3^- to NO_2^- by cadmium) has been previously applied to the spectrometric analysis of nitrate in water (Morris and Riley, 1963; Wood et al., 1967; Margeson et al., 1980). First, NO_3^- in water samples (50 mL) is reduced to NO_2^- in Teflon sealed batches through the addition of 1 g spongy cadmium (buffered with 1 M imidazole solution to pH 9) which is shaken overnight (horizontal) at 120 rpm. Freshwater samples require a sodium chloride concentration of 0.5 M. After separation of the spongy cadmium by centrifugation, the water samples are introduced into new vials. Schilman and Tepliyakov (2007) have proposed an alternate technique for this first reduction step which involves passing the sample through a cooperized granular-cadmium filled-column at a rate of 5 mL min^{-1} . Complete conversion of sample NO_3^- to NO_2^- , avoiding any isotopic fractionation, can be achieved within a few minutes. The samples are then filled into 60 mL vials, sealed with Teflon-lined septa and purged with N_2 or He to remove any N_2O . For the second reduction step, 2 mL 1:1 sodium azide (2 M)/acetic acid (20%) solution is injected into the sample to reduce NO_2^- to N_2O and N_2 . In order to remove any HN_3 , which is extremely toxic and volatile, 1 mL of 6 M NaOH is added. Positive aspects of the method include: (1) the preparation is less laboratory intensive (results can be available within one day) and consumables are inexpensive (cadmium can also be reused); (2) low concentrations ($0.5 \mu\text{mol L}^{-1}$) and small sample volumes can be analyzed; (3) no interferences with high concentrations and toxic substances; and (4) the possibility for

automation with high sample throughput. Negative aspects of the method include: (1) environmental and health protection measures due to the toxicity of the cadmium, sodium azide, and HN_3 ; (2) results require corrections for oxygen exchange and fractionation; and (3) the presence of NO_2^- in water samples can bias the isotopic composition of product N_2O and N_2 . As the “cadmium reduction method” is a new technique for $\delta^{15}\text{N}$ - and $\delta^{18}\text{O}$ - NO_3^- measurement, it requires further demonstration in the future.

11. Conclusions

Various potential NO_3^- sources have distinct $\delta^{15}\text{N}$ - and $\delta^{18}\text{O}$ - NO_3^- values and the dual isotope approach is a powerful tool to identify NO_3^- sources in contaminated water. Furthermore, incorporation of hydrochemistry may help to confirm NO_3^- source identification. However, complex fractionation processes and admixture from multiple NO_3^- sources usually make it difficult to correctly identify the NO_3^- source contribution. These problems may be eliminated by combining $\delta^{15}\text{N}$ - and $\delta^{18}\text{O}$ - NO_3^- with co-migrating discriminators of NO_3^- sources. For example, boron (B) isotopes, which are present as a minor or trace constituent in nearly all water types, are not affected by biogeochemical transformation processes. Meanwhile, reliable quantification and uncertainty assessment of a variety of NO_3^- source inputs via Bayesian framework estimation could provide powerful evidence for NO_3^- source contribution. In addition, the “ion-exchange method” and the “bacterial denitrification method” are frequently used for $\delta^{15}\text{N}$ - and $\delta^{18}\text{O}$ - NO_3^- measurement, but only a limited number of laboratories worldwide are capable to carry out these analyses. Hence, precise, accurate, efficient and broadly applicable isotope techniques need to be explored before NO_3^- source quantification using the isotope mixing model operating in a Bayesian framework can be implemented in large scale water monitoring programs.

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